



Electronics & ICT Academy

Indian Institute of Technology Guwahati

Offline Training Partner Under An Initiative of Ministry
of Electronics & Information Technology (Meity)



b Besant Technologies



Advanced certification In

DATA SCIENCE & AI

Includes



Gen AI



Transformers



LLMs & Prompt Engineering



100% Placement Assurance



Blended Classroom



7.5 Months 750 Hours



In Collaboration with



Electronics & ICT Academy Indian Institute of Technology Guwahati

Offline Training Partner Under An Initiative of Ministry of Electronics
& Information Technology (Meity)

About E&ICT Academy



E&ICT Academy is a joint initiative by the Ministry of Electronics & Information Technology (MeitY), Government of India and IIT Guwahati with the aim of training the forthcoming technical workforce to enhance employability in the country.

MeitY - Ministry of Electronics and Information Technology has sponsored a scheme entitled "Scheme of Financial Assistance for setting up of Electronics and ICT Academies". Electronics and ICT Academy would aim to provide specialised training to the faculties of Engineering, Arts, Commerce & Science colleges, Polytechnics etc, by developing state-of-the-art facilities. Academy has planned short term training programmes on fundamental and advanced topics in IT, Electronics & Communication, Product Design, Manufacturing with hands-on training and project work using latest software tools and systems

In addition, the Academy will conduct specialised/customised training programmes and research promotion workshops for corporate sector & educational institutions. It is envisaged that the Electronics and ICT Academy will become a central hub of activities on training, consultancy work and entrepreneurship programmes.



About Besant Technologies

Besant Technologies is a premier IT training institute in India, renowned for delivering high-quality, industry-aligned education tailored for both individuals and corporate clients aiming to excel in the IT sector. Since our inception, we have remained committed to bridging the gap between theoretical learning and real-world application, making us a trusted choice for aspiring IT professionals and engineering graduates. Whether you're a recent graduate seeking to launch your career or an experienced professional looking to upskill, Besant Technologies provides the tools, training, and placement support needed to thrive in today's dynamic IT landscape.

Member of
nasscom

Besant Technologies Integrity

1 Awards and Recognition

Besant Technologies has been consistently recognized as a top-performing IT training provider across South India. With 15+ years of industry presence, Besant has received accolades from reputed platforms like UrbanPro, Sulekha, and Just Dial for excellence in delivering quality training in Java, Python, Data Science, and Cloud technology.

2 Vast Training Experience

With over 1,00,000+ students trained across various technologies, Besant Technologies brings unmatched experience in delivering hands-on IT training. From fresh graduates to working professionals, our courses cater to a wide audience across technologies like Full Stack Development, AWS, Data Analytics, DevOps, AI, and more.

3 Vast Training Experience

Besant Technologies takes pride in its team of 150+ trainers, most of whom are working professionals from top MNCs like TCS, Infosys, Cognizant, Accenture, and Wipro. Each trainer brings 8–15 years of real-time project experience and is dedicated to providing learners with current, industry-relevant knowledge.

Exclusive Notes



A message from Principal Investigator, E&ICT Academy, IIT Guwahati

Dear Students,

As the Principal Investigator of E&ICT Academy, IIT Guwahati, I am pleased to endorse and collaborate with Besant Technologies our offline training center in Pune on their Advanced Certification in Cloud Computing & Devops.

At E&ICT Academy, IIT Guwahati, we are committed to promoting excellence in education and research in the field of Python Full Stack Development, Data Analytics, Data Science, Artificial Intelligence and Cloud Computing.

I wish you all the best in your educational journey with Besant Technologies and look forward to witnessing your growth and achievements in the exciting fields of Python Full Stack Development, Data Analytics, Data Science, Artificial Intelligence and Cloud Computing.

Warm Regards,

Prof. Gaurav Trivedi

E&ICT Academy, IIT Guwahati Certification

A Certification of course completion is given by E&ICT Academy, IIT Guwahati. This certification is supported by Ministry of Electronics and Information Technology (MeitY), Govt. of India.

Sample Certificate



In Collaboration with



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Indian Institute of Technology Guwahati

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Welcome to the **DATA SCIENCE** Master Program

Master the Art of Data-Driven
Decisions



In today's fast-growing digital world, data is more than just numbers . it's the key to making smart business decisions. At **Besant Technologies**, we believe in empowering students with real-world skills that are in demand across industries.

Our **Data Science Master Program** is designed to help you become job-ready by mastering the tools and techniques used by data scientists across the globe. Whether you're a student, working professional, or someone looking for a career switch, this course will give you the knowledge and hands-on experience needed to stand out in the data-driven world.

Why Data Science?



One of the most in-demand careers in the world



High-paying job opportunities across all industries



Core of AI, Machine Learning, and modern technology



Enables smarter, data-driven decisions

Our Futures



Real-Time Case Studies



Own Prepared Materials



Placement Oriented Courses



Experienced MNC Professionals



Syllabus Based on Companies



Corporate Style Training



Industry-Recognized Certifications



Hands-On Practical Sessions



CAREER OPTION DATA SCIENCE

Our Credentials



**10 Million+ Aspiring
Active Students**



**1,000+ Industry-expert
Instructors**



400+ Hiring Partners



155+ Countries' Learners



**500+ Corporates
Upskilled**



55% Average Salary Hike

"In God we trust; all others must bring data."

William Edwards Deming

Why Choose Data Science?

Data is the New Power!

In today's world, every business — from startups to tech giants — depends on data to grow, innovate, and succeed. Data Science is at the heart of this revolution, making it one of the fastest-growing and most exciting career paths.



Top Reasons to Choose a Career in Data Science



Massive Demand, Unlimited Opportunities

Companies across industries like IT, Healthcare, Finance, Retail, and Entertainment are hiring data scientists in huge numbers.



Versatile Career Options

With Data Science skills, you can work as a Data Analyst, Machine Learning Engineer, Data Engineer, Business Intelligence Analyst, and many more.



Work on Real-World Challenges

Solve real business problems, predict market trends, improve healthcare systems, and create smarter technologies.



High Salary Packages

Data Science professionals are among the highest-paid experts globally, with endless opportunities for career growth.



What Makes Us Different?

- ✓ **Industry-Expert Trainers:** Our trainers are experienced professionals working in top MNCs. You'll learn the latest tools, technologies, and real-world practices.
- ✓ **Hands-on Learning Approach:** We believe in learning by doing. Every module includes practical labs, mini-projects, and case studies to strengthen your knowledge.
- ✓ **Placement Support:** Our dedicated placement team helps with interview training, resume preparation, and connects you with top recruiters.
- ✓ **Real-Time Projects & Capstone:** Work on industry-level projects that reflect real business challenges and boost your portfolio.
- ✓ **Flexible Learning Modes:** Choose classroom or online training as per your comfort. Learn at your own pace with recorded sessions available.
- ✓ **Affordable Fees + EMI Options:** Quality education shouldn't be expensive. Our pricing is student-friendly, with easy installment plans.
- ✓ **Globally Recognized Certification:** Get a professional certificate that validates your skills and boosts your job profile.





PROGRAM OVERVIEW

MASTER DATA SCIENCE WITH INDUSTRY EXPERTS

Become a Job-Ready Data Science Professional

Our **Data Science Master Program** is built by experienced industry professionals to ensure that students learn exactly what today's companies expect from data scientists. This program combines theoretical **knowledge, hands-on practice, and industry insights** to give you a 360° learning experience.

Whether you're a fresher, working professional, or someone planning a career shift, this program is your gateway to mastering Data Science from scratch. guided by the people who live and breathe it daily.



LEARN FROM THE BEST IN THE INDUSTRY

Our trainers are working professionals from top MNCs, bringing real-time use cases, project experience, and industry trends directly into the classroom. This means:



- ✓ You learn current tools and technologies
- ✓ You gain practical skills, not just theory
- ✓ You get insider tips on cracking interviews and excelling at work

Course Designed & Delivered by Experts from



COURSE PREREQUISITE – WHO CAN JOIN?

No Prior Experience? No Problem!

Our **Data Science Master Program** is designed for learners from all educational and professional backgrounds. Whether you're from IT, commerce, engineering, or arts. you can confidently start your journey with us.

No Programming Background?

Don't worry!

We start from the basics of Python programming, and gradually take you to advanced levels. You'll be trained step-by-step — with examples, practice, and full trainer support.

➞ You are eligible if you

- Have basic computer knowledge
- Are comfortable with high school level math
- Have an interest in data, logic, or problem-solving
- Want to build a career in tech or analytics

➞ Ideal for

- Final-year students & fresh graduates
- Working professionals planning a career switch
- Business analysts & non-technical employees upskilling
- Anyone passionate about building a career in Data Science, AI, or Machine Learning

WHAT YOU'LL LEARN?

A Complete Roadmap to Becoming a Data Scientist

Our **Data Science Master Program** is structured to take you from beginner level to expert, covering every essential skill that top companies expect. You'll gain both technical knowledge and practical experience through hands-on training.

Core Concepts Covered

- ✓ **Python for Data Science:** Learn Python from scratch, including data types, functions, loops, libraries like Pandas and NumPy.
- ✓ **Statistics & Probability:** Master descriptive and inferential statistics, probability concepts, and hypothesis testing.
- ✓ **SQL & Databases:** Work with structured data using SQL – query writing, joins, subqueries, and data filtering.
- ✓ **Data Wrangling & Visualization:** Clean messy data and create powerful dashboards using Matplotlib, Seaborn, and Power BI/Tableau
- ✓ **Machine Learning:** Dive deep into supervised & unsupervised algorithms: Linear Regression, Decision Trees, K-Means, Random Forest, SVM, and more.
- ✓ **Deep Learning (Basics):** Get introduced to neural networks and deep learning using TensorFlow/Keras.
- ✓ **Model Deployment:** Learn how to take your models live using Flask, Streamlit, or cloud platforms.

Tools & Technologies You'll Master:

- Python
- Pandas, NumPy, Scikit-learn
- SQL
- Power BI & Tableau
- Gen AI
- TensorFlow & Keras
- GitHub
- Jupyter Notebook
- Google Colab
- Excel and Advanced Excel
- Statistics
- Deep Learning
- Machine Learning
- AI



GLOBAL DATA SCIENCE MASTER COURSE MARKET OVERVIEW

Data Science is the Fuel of the Future

In the digital age, data is more valuable than oil. Companies around the world are investing heavily in data-driven technologies to improve decision-making, automation, customer experience, and innovation.

This has created an explosive demand for data science professionals across industries from finance and healthcare to e-commerce, logistics, and entertainment.

MARKET HIGHLIGHTS

- ✓ Over 11 million global job openings in Data Science expected by 2026 (Source: World Economic Forum)
- ✓ Data Science industry projected to grow to \$378 billion by 2030
- ✓ Top recruiters: Google, Amazon, Microsoft, IBM, Accenture, Infosys, Flipkart, and more
- ✓ Average Salary:
 - India: ₹8–15 LPA (Entry to Mid-level)
 - USA: \$100K+ (Entry-level)
 - UK, Germany, Canada: High demand with visa-sponsored jobs.

WHY NOW IS THE BEST TIME TO START?

- ✓ Rapid adoption of AI & Machine Learning.
- ✓ Massive growth in cloud computing & big data analytics.
- ✓ Demand is outpacing supply – skilled professionals are highly valued
- ✓ Companies want professionals who are certified, trained, and project-ready.

MEET THE BATCH

LEARN WITH A DIVERSE COMMUNITY

Our **Data Science Master Program** attracts learners from various industries, creating a dynamic and engaging learning environment. You will collaborate with IT professionals, developers, system administrators, and career changers who bring unique insights and experiences to the program.

Who Joins This Program?



Working Professionals – IT engineers, software developers, and DevOps specialists looking to upskill.



Career Changers – Professionals from non-IT backgrounds transitioning into cloud computing roles.



Fresh Graduates & Students – Those seeking industry-relevant cloud skills and certifications.



Freelancers & Entrepreneurs – Individuals looking to leverage cloud computing for their projects or businesses.

SOME OF THEIR CURRENT EMPLOYERS INCLUDE



Data Science Placement Outcomes

Name	Role	Y.O.P	Qualification	CTC
 Kartik Kumar	Data Scientist	2023	(NON IT TO IT)	12 LPA
 Yamini	Data Science Engineer	2024	(NON IT TO IT)	6 LPA
 Yakshitha Valluri	Associate Admin Security	2023	B.Tech	5 LPA
 Dhara Bala	Data Scientist	2023	(NON IT TO IT)	3.5 LPA
 Baisakhi Nayak	Associate Data Operation	2023	B.Sc(Maths)	4.7 LPA
 Samuel	Data Scientist	2017 (7 Years career break to IT)	B.E(Mech)	4 LPA

Data Science Placement Outcomes

Name	Role	Y.O.P	Qualification	CTC
 Arjun Gunaseelan	Data Scientist	2022	B.Tech	9 LPA
 Pandi Selvam	Data Science Engineer	2023	MCA	2.4 LPA
 Kothala Ganesh	Data Engineer	2023	B.Tech(ECE)	4 LPA
 Teja	Data Scientist	2020	MBA (NON IT TO IT)	3 LPA
 Arivumathi	Data Scientist	2023	M.Sc(Data Science)	3.5 LPA
 Sanjai	Power BI Developer	2019 (4 Years career break to IT)	MCA	3 LPA
 Aravinth	System Administrator	2017 (5 Years career break to IT)	B.E(Mech) (NON IT TO IT)	4.5 LPA
 Vikas	Operation Associate	2023	MBA	3.6 LPA

PROJECT INNOVATION LAB

Our Project Innovation Lab is designed to provide hands-on, real-world experience in automation testing. It serves as a practical learning environment where learners can apply their skills to real-time projects, industry scenarios, and automation frameworks.

- ✔ **Real-World Cloud Deployments:** Build, test, and deploy cloud-based applications in a hands-on environment.
- ✔ **Portfolio Building:** Showcase projects to enhance job opportunities.
- ✔ **Collaboration & Peer Learning:** Work with a diverse group of learners and mentors to tackle complex cloud problems.
- ✔ **Live Industry Projects:** Work on projects modeled after real business scenarios, such as cloud migration, serverless applications, and multi-cloud solutions.
- ✔ **Experiment with Emerging Tech:** Explore AI, IoT, Blockchain, and Big Data within cloud environments.
- ✔ **Hackathons & Challenges:** Participate in cloud hackathons, problem-solving challenges, and innovation sprints to sharpen your skills.

BESANT TECHNOLOGIES CAREER SERVICES



**7 Months Live
Instruction**



**400+ Hiring
Partners**



**500+ Job
Shares**

What Makes Us Best

✓ **Resume Building**

Crafting a targeted resume that highlights your technical skills, relevant experience, and achievements.

✓ **Networking Assistance**

Offering guidance on leveraging social media platforms like LinkedIn for professional networking.

✓ **Career Counseling and Guidance**

Offering one-on-one counseling sessions to discuss career goals, aspirations, and challenges.

✓ **Job Search Strategies**

Offering guidance on effective job search techniques, including networking, online applications, and recruiter outreach.

✓ **Skill Development and Training**

Tailoring skill development plans based on current industry trends and job market demand.

“

“The world is moving towards data-first decisions. Join this revolution and build a borderless career in Data Science.”

OUR DATA SCIENCE STUDENTS FEEDBACK

M**Manikandan | IT Professional**

I have completed the Data Science course from Besant Technologies recently. My trainer, Sundar Mohan, has maintained a very good standard of the course and has taught us very well along with the practical knowledge in different scenarios. He practical knowledge in different scenarios. He patiently took our problems and doubts and solved them. I am grateful to my trainer and Besant Technologies.

R**Raja | IT Professional**

I never thought I could be a part of the IT world, given that I was not from a science background. Thankfully, Besant Technologies doesn't need you to have any special educational background. The faculty members here taught us data science training concepts so well, that never for a moment I felt I was no way related to technology. Now, I have joined an IT MNC. And working in a good position.

G**Ganga | IT Professional**

I am an ex Besant Technologies student. Although I liked everything about the coaching center, I especially loved the way the data science training trainers imparts the course lessons. They not just emphasize theories but also practical. With regular practical training, I have learned a lot about the subject and its applicability. This has immensely helped in my job.

K**Karunakaran | IT Professional**

I always wanted to do something in the field of data science/Analytics. But I was not being able to find out a good training center. Then an ex Besant Technologies student told me about this institute. Besant Technologies has trained me so well in data science training that I was reined in by the first company that interviewed me. I feel confident and blessed today. All thanks to Besant Technologies!

R**Ram Kumar | IT Professional**

I was a Junior data scientist. It was my sheer interest in data science that compelled me to take up the data science training at Besant Technologies. Since I already had a little knowledge of the subject, it was not really extremely difficult for me to have a grip on the course. But the training was imparted by the experienced teachers of Besant so effectively that even those fellow candidates who had no background in data science showed progress pretty fast.

PROGRAM HIGHLIGHTS

- ✓ 100+ hours of Practical Learning
- ✓ Lifetime access to class recordings
- ✓ One to one training
- ✓ Career Placement Support
- ✓ 24*7 Support
- ✓ Unlimited Interviews
- ✓ 50+ Live sessions across 7 months
- ✓ Gain insights from industry experts
- ✓ 3 industry projects with Multiple case studies
- ✓ Course Delivered by Certified Professionals
- ✓ Resume Preparation and LinkedIn Profile Review
- ✓ 100% Placement Oriented Classes



DATA SCIENCE

MASTER TRAINING

- Python
- Statistics
- Data Science
- Machine learning
- Deep Learning
- Artificial Intelligence
- Advanced ML & Generative AI: Transformers, LLMs & Prompt Engineering
- Excel and Advanced Excel
- SQL
- Power BI
- Tableau

Add-on course with this Package

👉 **Project Work**

👉 **Aptitude**

Average Salaries: ₹ 6L - ₹ 17L Per Annum

Hiring Companies: TCS, Accenture, Amazon ,
CISCO, Capgemini Etc

PYTHON TRAINING

INTRODUCTION TO PYTHON

Objective:

In this module, you will get a basic understanding of python programming, Virtual environment, Package manager, version differences of python programming and detailed knowledge about the python installation and environment setup for working with python.

Hands-on-Exercise:

Topics :

- ✓ Python programming history & features
- ✓ Python compiler and IDE installation
- ✓ Virtual Environment
- ✓ Pip – Package Manager

Hands-on

How to set up an environment for python development.

Basics of Python:

Objective

In this module, you will get a basic understanding of python Syntax and a detailed understanding of Input/Output [I/O] operations, Variables, Operators Datatypes and Data structure.

Topics :

- ✓ Python Syntax Overview, Indentation, comments
- ✓ Variable declaration
- ✓ Datatypes and data structure
 - Primitive
 - Non-primitive
- ✓ Pip – Package Manager

Hands-on

How to write a basic Python code, variable declaration datatypes, and operators usage.

Program flow/ Data flow of Python:

Objective

In this module, you will get a detailed understanding of conditional statements, looping, control statements of python.

Topics :

- ✓ Conditional Statements
 - if statement
 - if ... else statement
 - if ... elif... else statement
- ✓ Looping
 - for loop
 - for with else statement

- while loop
- while with else statement
- ✓ **Control Statements**
 - break
 - Continue
 - pass

Hands-on

How to write a basic Python code, variable declaration datatypes, and operators usage.

Function in Python:

Objective

In this module, you will get a detailed understanding of conditional statements, looping, control statements of python.

Topics :

- ✓ Syntax of Function
- ✓ Function with *args & **kwargs
- ✓ Scope of variables
- ✓ Lambda function with map, filter, reduce method
- ✓ DocString
- ✓ Modules and standard Modules

Hands-on

How to create and use functions and different types of argument, lambda function, and modules in python.

File Handling in Python:

Objective

In this module, you will get a detailed understanding of file concepts like create, read, write, update, delete operation of file with python.

Topics :

- ✓ File Opening modes
- ✓ Context Manager in python
 - Open
 - Create
 - Read
 - Write
 - Update
 - Delete

Hands-on

How to create and use file operation in python.

Exception Handling in Python:

Objective

In this module, you will get a detailed understanding of exception handling and creating your own exception classes in python

Topics :

- ✓ Types of Errors in python
- ✓ Exception handling with
 - try ... except
 - try ... except... finally
 - try ... except... else
- ✓ Multiple Exception
- ✓ Raising Exception
- ✓ User-defined Exception

Hands-on

How to create and use built-in and user-defined exception handling in python.

Oops in Python:

Objective

In this module, you will get a detailed understanding of Object-Oriented Programming Concepts in python.

Topics :

- ✓ Oops Concepts with programming syntax
 - Class
 - Object
 - Polymorphism
 - Encapsulation
 - Inheritance

✓ **Types of Methods in python**

Hands-on

How to use Object-oriented programming concepts in python.

Core Concepts in Python:

Objective

In this module, you will get a detailed understanding of iterators, generators, decorators, in python.

Topics :

- Iterator
- Generator
- Decorator

Hands-on

How to use core concepts and application of core concepts in python.

Comprehension in Python:

Objective

In this module, you will get a detailed understanding of List, Dictionary comprehension and specialized sorts in python.

Topics :

- ✓ **Comprehensions**
- List
 - Nested List

- o if statement
- o if ... else statement
- o Nested if ... else statement
- Dictionary
- Sorting
 - o List
 - o Dictionary

Hands-on

How to use comprehensions and sorting in python.

Thread and DateTime in Python:

Objective

In this module, you will get a detailed understanding of multi-threading concepts, datetime module along with sleep and execution of code in python.

Topics :

- ✓ Terms in threading
 - process
 - thread
 - multithreading
 - Time complexity
- ✓ Thread Life cycle
- ✓ Programming with Threading & Multithreading
- ✓ Synchronization

- ✓ Sleep and execution time of code
- ✓ DateTime module

Hands-on

How to use threading and data time concepts in python.

Advanced data Structure/ collections in Python:

Objective

In this module, you will get a detailed understanding of advanced data structures in python.

Topics :

- ✓ Deque
- ✓ namedtuple
- ✓ ChainMap
- ✓ Counter
- ✓ Ordered Dictionary
- ✓ Default Dictionary

Hands-on

How to use advanced data structure in python.

MySQL with Python:

Objective

In this module, you will get a detailed understanding of SQL statements and database connection along with CRUD operation using python.

Topics :

- ✓ SQL statements & Operations
 - Create
 - Read
 - Update
 - Delete
- ✓ Python – SQL connector package installation
- ✓ Python with CRUD Operations
- ✓ Commit & Rollback
- ✓ SQL Related Exception Handling

Hands-on

How to use and manipulate data in a database using python.

Network programming with Python:

Objective

In this module, you will get a detailed understanding of Network programming, Client & server concepts with python.

Topics :

- ✓ Terms and Basics of network programming
- ✓ The architecture of data transmission between sender and receiver using python
- ✓ Getting data from the remote server
- ✓ Client & Server-side programming

Hands-on

How to use and networking module of python, data transmission between client to server and server to client python.

Regular Expression with Python:

Objective

In this module, you will get a detailed understanding of writing Regex with python.

Topics :

- ✓ Regex Syntax
 - Quantifiers
 - Metacharacters
 - Special Sequences
 - Sets
- ✓ Python re module
- ✓ Methods with regex usage

Hands-on

How to use and write regex in python.

GUI programming with Python:

Objective

In this module, you will get a detailed understanding of developing the GUI application using the PyQt5 module with python.

Topics :

- ✓ Introduction
- ✓ Components and Events
- ✓ An Example GUI
- ✓ Widgets
- ✓ Layout Management
- ✓ Signals & Slots
- ✓ QMessageBox, QDialo
- ✓ Database Handling

Hands-on

How to develop a GUI application with PyQt5 and python.

API access with Python:

Objective

In this module, you will get a detailed understanding of accessing open APIs using python.

Topics :

- ✓ Google Text to Speech
- ✓ Google Speech to Text
- ✓ OpenWeatherMap

Hands-on

How to use open APIs using python.

Data Science with Python:

Objective

In this module, you will get a Basic understanding of data science modules in python.

Topics :

- ✓ Pandas – Series and Dataframe
- ✓ Numpy
- ✓ Matplotlib

Hands-on

How to use data science modules of python.

The project with Python:

Objective

In this module, you are going to develop an application for own scenario.

Topics :

- ✓ Creating own application with any one of the frameworks
 - Django App
 - PyQt5 App
 - Console oriented Core app

STATISTICS BASICS

- ✓ **Central Tendency**
 - Mean
 - Median
 - Mode
 - Skewness
 - Normal Distribution
- ✓ **Probability Basics**
 - What does it mean by probability?
 - Types of Probability
 - ODDS Ratio
- ✓ **Standard Deviation**
 - Data deviation & distribution
 - Variance
- ✓ **Bias variance Tradeoff**
 - Underfitting
 - Overfitting
- ✓ **Distance metrics**
 - Euclidean Distance
 - Manhattan Distance
- ✓ **Outlier analysis**
 - What is an Outlier?
 - Inter Quartile Range

- Box & whisker plot
- Upper Whisker
- Lower Whisker
- Scatter plot
- Cook's Distance
- ✓ Missing Value treatment
 - What is NA?
 - Central Imputation
 - KNN imputation
 - Dummification
- ✓ Correlation
 - Pearson correlation
 - positive & Negative correlation

Hands-on

- ✓ Compute probability in a situation where there are equally-likely outcomes
- ✓ Apply concepts to cards and dice
- ✓ Compute the probability of two independent events both occurring
- ✓ Compute the probability of either of two independent events occurring
- ✓ Do problems that involve conditional probabilities
- ✓ Calculate the probability of two independent events occurring
- ✓ List all permutations and combinations
- ✓ Apply formulas for permutations and combinations

Module 8: Error Metrics

- ✓ **Classification**
 - Confusion Matrix
 - Precision
 - Recall
 - Specificity
 - F1 Score
- ✓ **Regression**
 - MSE
 - RMSE
 - MAPE

Hands-on-Exercise:

- ✓ State why the z' transformation is necessary
- ✓ Compute the standard error of z
- ✓ Compute a confidence interval on p The computation of a confidence interval
- ✓ Estimate the population proportion from sample proportions
- ✓ Apply the correction for continuity

MACHINE LEARNING

Supervised Learning

- ✓ Linear Regression
 - Linear Equation
 - Slope
 - Intercept
 - R square value
- ✓ Logistic regression
 - ODDS ratio
 - Probability of success
 - Probability of failure Bias Variance Tradeoff
 - ROC curve
 - Bias Variance Tradeoff

Hands-on

We've reviewed the main ways to approach the problem of modeling data using simple and definite'

- ✓ Unsupervised Learning (Duration-4hrs)
 - K-Means
 - K-Means ++
 - Hierarchical Clustering

- ✓ **SVM (Duration-2hrs)**
 - Support Vectors
 - Hyperplanes
 - 2-D Case
 - Linear Hyperplane
- ✓ **SVM Kernal (Duration-2hrs)**
 - Linear
 - Radial
 - polynomial
- ✓ **Other Machine Learning algorithms (Duration-10hrs)**
 - K – Nearest Neighbour
 - Naïve Bayes Classifier
 - Decision Tree – CART
 - Decision Tree – C50
 - Random Forest

Hands-on

- ✓ We have covered the simplest but still very practical machine learning models in an eminently practical way to get us started on the complexity
- ✓ where we will cover several regression techniques, it will be time to go and solve a new type of problem that we have not worked on, even if it's possible to solve the problem with clustering methods (regression), using new mathematical tools for approximating unknown values.
- ✓ In it, we will model past data using mathematical functions, and try to model new output based on those modelling

ARTIFICIAL INTELLIGENCE

✓ Module 1: AI Introduction (Duration-9hrs)

- Perceptron
- Multi-Layer perceptron
- Markov Decision Process
- Logical Agent & First Order Logic
- AL Applications

Deep Learning

✓ Module 1: Deep Learning Algorithms (Duration-10hrs)

- CNN – Convolutional Neural Network
- RNN – Recurrent Neural Network
- ANN – Artificial Neural Network

Hands-on-Exercise:

- We took a very important step towards solving complex problems together by means of implementing our first neural
- Now, the following architectures will have familiar elements, and we will be able to extrapolate the knowledge acquired on this chapter, to novel

✓ Introduction to NLP (Duration-5hrs)

- Text Pre-processing
- Noise Removal

- Lexicon Normalization
 - Lemmatization
 - Stemming
 - Object Standardization
- ✓ **Text to Features (Feature Engineering) (Duration-5hrs)**
- Syntactical Parsing
 - Dependency Grammar
 - Part of Speech Tagging
 - Entity Parsing
 - Named Entity Recognition
 - Topic Modelling
 - N-Grams
 - TF – IDF
 - Frequency / Density Features
 - Word Embedding's
- ✓ **Tasks of NLP (Duration-2hrs)**
- Text Classification
 - Text Matching
 - Levenshtein Distance
 - Phonetic Matching
 - Flexible String Matching

Hands-on

- ✓ **Provided, you will even be able to create new customized**

- ✓ As our models won't be enough to solve very complex problems, in the following chapter, our scope will expand even more, adding the important dimension of time to the set of elements included in our generalization.

Project Works

- ✓ **Project 1: Board Game Review Prediction**
 - To perform a Linear regression
 - Analysis by predicting the average reviews in a board game
- ✓ **Project 2 :Credit Card Fraud Detection**
 - TO focus on Anomaly Detection by using probability densities to detect credit card fraud
- ✓ **Project 3: Stock Market Clustering**
 - Learn how to use the K-means clustering
 - To find related companies by finding correlations among stock market movements over a given time span
- ✓ **Project 4: Getting Started with Natural Language Processing**
 - Will focus on Natural Language Processing (NLP) methodology, such as tokenizing words and sentences, part of speech identification and tagging, and phrase
- ✓ **Project 5: Obtaining Near State-of-the-Art Performance on Object Recognition**
 - Using Deep Learning – In this project, will use the CIFAR-10 object recognition dataset as a benchmark to implement a recently published deep neural

- ✓ **Project 6: Image Super Resolution with the SRCNN – Learn how to implement & use**
 - Tensorflow version of the Super Resolution Convolutional Neural Network (SRCNN) for improving image
- ✓ **Project 7: Natural Language Processing: Text Classification**
 - An advanced approach to Natural Language Processing by solving a text classification task using multiple classification
- ✓ **Project 8: K-Means Clustering For Image Analysis**
 - Use K-Means clustering in an unsupervised learning method to analyze and classify 28 x 28 pixel images from the MNIST
- ✓ **Project 9: Data Compression & Visualization Using Principal Component Analysis**
 - This project will show you how to compress our Iris dataset into a 2D feature set and how to visualize it through a normal x-y plot using k-means clustering

EXCEL AND ADVANCED EXCEL COURSE

✓ Module 1: Basic Excel (Duration: 10Hrs)

- Text to Columns
- Concatenate
- The Concatenate Function
- The Right Function with Concatenation
- Absolute Cell References
- Data Validation
- Time and Date Calculations
- Conditional Formatting
- Exploring Styles and Clearing Formatting
- Using Conditional Formatting to Hide Cells
- Using the IF Function
- Changing the “Value if false” Condition to Text
- Pivot Tables
- Creating a Pivot Table
- Specifying PivotTable Data
- Changing a PivotTables Calculation
- Filtering and Sorting a PivotTable
- Creating a PivotChart
- Grouping Items
- Updating a PivotTable
- Formatting a PivotTable

- Using Slicers
- Charts
- Creating a Simple Chart
- Charting Non-Adjacent Cells
- Creating a Chart Using the Chart Wizard
- Modifying Charts
- Moving an Embedded Chart
- Sizing an Embedded Chart
- Changing the Chart Type
- Chart Types
- Changing the Way Data is Displayed
- Moving the Legend
- Formatting Charts
- Adding Chart Items
- Formatting All Text
- Formatting and Aligning Numbers
- Formatting the Plot Area
- Formatting Data Markers
- Pie Charts
- Creating a Pie Chart
- Moving the Pie Chart to its Own Sheet
- Adding Data Labels
- Exploding a Slice of a Pie Chart
- Data Analysis – Overview

- types of Data Analysis
- Data Analysis Process
- Working with Range Names
- Copying Name using Formula Autocomplete
- Range Name Syntax Rules
- Creating Range Names
- Creating Names for Constants
- Managing Names
- Scope of a Name
- Editing Names
- Applying Names
- Using Names in a Formula
- Viewing Names in a Workbook
- Copying Formulas with Names
- Difference between Tables and Ranges
- Create Table
- Table Name
- Managing Names in a Table
- Table Headers replacing Column Letters
- Propagation of a Formula in a Table
- Resize Table
- Remove Duplicates
- Convert to Range
- Table Style Options

- Table Styles
- Cleaning Data with Text Functions
- Removing Unwanted Characters from Text
- Extracting Data Values from Text
- Formatting Data with Text Functions

Advance Excel (Duration: 15Hrs)

✓ Module 2: Date Formats (Duration: 2Hrs)

- Converting Dates in Serial Format to Month-Day-Year Format
- Converting Dates in Month-Day-Year Format to Serial Format
- Obtaining Today's Date
- Finding a Workday after Specified Days
- Customizing the Definition of a Weekend
- Number of Workdays between two given dates
- Extracting Year, Month, Day from Date
- Extracting Day of the Week from Date
- Obtaining Date from Year, Month and Day
- Calculating Years, Months and Days between two dates

✓ Module 3: Conditional Formatting (Duration: 2Hrs)

- Highlight Cells Rules
- Top / Bottom Rules
- Data Bars
- Color Scales
- Icon Sets

- New Rule
- Clear Rules
- Manage Rules

✓ **Module 4: Sorting (Duration: 2Hrs)**

- Sort by Text
- Sort by Numbers
- Sort by Dates or Times
- Sort by Cell Color
- Sort by Font Color
- Sort by Cell Icon
- Sort by a Custom List
- Sort by Rows
- Sort by more than one Column or Row

✓ **Module 5: Filtering (Duration: 2Hrs)**

- Filter by Selected Values
- Filter by Text
- Filter by Date
- Data Analysis with Excel
- Filter by Numbers
- Filter by Cell Color
- Filter by Font Color
- Filter by Cell Icon
- Clear Filter

- Advanced Filtering
- Filter Using Slicers

✓ **Module 6: Other Topics (Duration: 2Hrs)**

- Subtotals with Ranges
- Subtotals
- Nested Subtotals
- Quick Analysis
- Quick Analysis with TOTALS
- Sum
- Average
- Count
- %Total
- Running Total
- Sum of Columns

✓ **Module 7: Lookup Functions (Duration: 3Hrs)**

- Using the VLOOKUP Function
- Using VLOOKUP Function with range lookup TRUE
- Using VLOOKUP Function with range lookup FALSE
- Using the HLOOKUP Function
- Using HLOOKUP Function with range lookup FALSE
- Using HLOOKUP Function with range lookup TRUE
- Using the INDEX Function
- Using MATCH Function

✓ **Module 8: Pivoting (Duration: 2Hrs)**

- PivotTables
- Creating PivotTable
- Recommended PivotTables
- PivotTable Fields
- PivotTable Areas
- Nesting in the PivotTable
- Filters
- Slicers
- Summarizing Values by other Calculations
- PivotTable Tools
- Using Pictures in Column Charts
- Band Chart
- Thermometer Chart
- Gantt Chart
- Waterfall Chart
- Sparklines
- Pivot Charts
- PivotChart from PivotTable
- PivotChart without a PivotTable
- Working with Multiple Sheets
- Multiple Worksheets with same Structure
- Creating a Formula across Multiple Worksheets
- Summarizing Data in Multiple Worksheets
- What-If Analysis

SQL TRAINING

- ✓ **Introduction to Oracle Database**
 - List the features of Oracle Database 11g
 - Discuss the basic design, theoretical, and physical aspects of a relational database
 - Categorize the different types of SQL statements
 - Describe the data set used by the course
 - Log on to the database using SQL Developer environment
 - Save queries to files and use script files in SQL Developer
- ✓ **Retrieve Data using the SQL SELECT Statement**
 - List the capabilities of SQL SELECT statements
 - Generate a report of data from the output of a basic SELECT statement
 - Select All Columns
 - Select Specific Columns
 - Use Column Heading Defaults
 - Use Arithmetic Operators
 - Understand Operator Precedence
 - Learn the DESCRIBE command to display the table structure
- ✓ **Learn to Restrict and Sort Data**
 - Write queries that contain a WHERE clause to limit the output retrieved

- List the comparison operators and logical operators that are used in a WHERE clause
 - Describe the rules of precedence for comparison and logical operators
 - Use character string literals in the WHERE clause
 - Write queries that contain an ORDER BY clause to sort the output of a SELECT statement
 - Sort output in descending and ascending order
- ✓ Usage of Single-Row Functions to Customize Output
- Describe the differences between single row and multiple row functions
 - Manipulate strings with character function in the SELECT and WHERE clauses
 - Manipulate numbers with the ROUND, TRUNC, and MOD functions
 - Perform arithmetic with date data
 - Manipulate dates with the DATE functions
- ✓ Invoke Conversion Functions and Conditional Expressions
- Describe implicit and explicit data type conversion
 - Use the TO_CHAR, TO_NUMBER, and TO_DATE conversion functions
 - Nest multiple functions
 - Apply the NVL, NULLIF, and COALESCE functions to data
 - Use conditional IF THEN ELSE logic in a SELECT statement

- ✓ **Aggregate Data Using the Group Functions**
 - Use the aggregation functions in SELECT statements to produce meaningful reports
 - Divide the data in groups by using the GROUP BY clause
 - Exclude groups of data by using the HAVING clause
- ✓ **Display Data From Multiple Tables Using Joins**
 - Write SELECT statements to access data from more than one table
 - View data that generally does not meet a join condition by using outer joins
 - Join a table by using a self join
- ✓ **Use Sub-queries to Solve Queries**
 - Describe the types of problem that sub-queries can solve
 - Define sub-queries
 - List the types of sub-queries
 - Write single-row and multiple-row sub-queries
- ✓ **The SET Operators**
 - Describe the SET operators
 - Use a SET operator to combine multiple queries into a single query
 - Control the order of rows returned
- ✓ **Data Manipulation Statements**
 - Describe each DML statement
 - Insert rows into a table

- Change rows in a table by the UPDATE statement
- Delete rows from a table with the DELETE statement
- Save and discard changes with the COMMIT and ROLLBACK statements
- Explain read consistency
- ✓ **Use of DDL Statements to Create and Manage Tables**
 - Categorize the main database objects
 - Review the table structure
 - List the data types available for columns
 - Create a simple table
 - Decipher how constraints can be created at table creation
 - Describe how schema objects work
- ✓ **Other Schema Objects**
 - Create a simple and complex view
 - Retrieve data from views
 - Create, maintain, and use sequences
 - Create and maintain indexes
 - Create private and public synonyms
- ✓ **Control User Access**
 - Differentiate system privileges from object privileges
 - Create Users
 - Grant System Privileges
 - Create and Grant Privileges to a Role

- Change Your Password
- Grant Object Privileges
- How to pass on privileges?
- Revoke Object Privileges

✓ Management of Schema Objects

- Add, Modify and Drop a Column
- Add, Drop and Defer a Constraint
- How to enable and disable a Constraint?
- Create and Remove Indexes
- Create a Function-Based Index
- Perform Flashback Operations
- Create an External Table by Using ORACLE_LOADER and by Using ORACLE_DATAPUMP
- Query External Tables

✓ Manage Objects with Data Dictionary Views

- Explain the data dictionary
- Use the Dictionary Views
- USER_OBJECTS and ALL_OBJECTS Views
- Table and Column Information
- Query the dictionary views for constraint information
- Query the dictionary views for view, sequence, index and synonym information
- Add a comment to a table
- Query the dictionary views for comment information

- ✓ **Manipulate Large Data Sets**
 - Use Subqueries to Manipulate Data
 - Retrieve Data Using a Subquery as Source
 - Insert Using a Subquery as a Target
 - Usage of the WITH CHECK OPTION Keyword on DML Statements
 - List the types of Multitable INSERT Statements
 - Use Multitable INSERT Statements
 - Merge rows in a table
 - Track Changes in Data over a period of time
- ✓ **Data Management in Different Time Zones**
 - Time Zones
 - CURRENT_DATE, CURRENT_TIMESTAMP, and LOCALTIMESTAMP
 - Compare Date and Time in a Session's Time Zone
 - DBTIMEZONE and SESSIONTIMEZONE
 - Difference between DATE and TIMESTAMP
 - INTERVAL Data Types
 - Use EXTRACT, TZ_OFFSET and FROM_TZ
 - Invoke TO_TIMESTAMP, TO_YMINTERVAL and TO_DSINTERVAL
- ✓ **Retrieve Data Using Sub-queries**
 - Multiple-Column Subqueries
 - Pairwise and Nonpairwise Comparison
 - Scalar Subquery Expressions

- Solve problems with Correlated Subqueries
- Update and Delete Rows Using Correlated Subqueries
- The EXISTS and NOT EXISTS operators
- Invoke the WITH clause
- The Recursive WITH clause

✓ Regular Expression Support

- Use the Regular Expressions Functions and Conditions in SQL
- Use Meta Characters with Regular Expressions
- Perform a Basic Search using the REGEXP_LIKE function
- Find patterns using the REGEXP_INSTR function
- Extract Substrings using the REGEXP_SUBSTR function
- Replace Patterns Using the REGEXP_REPLACE function
- Usage of Sub-Expressions with Regular Expression Support
- Implement the REGEXP_COUNT function



POWER BI TRAINING

Introduction to Power BI

Learning Objectives

In this module, you will be introduced to what Power BI is, why to choose power BI, its building blocks, other various fundamental concepts of Power BI

Topics Covered:

- ✓ What is Power BI?
- ✓ Why Power BI?
- ✓ Benefits of Power BI
- ✓ Building Blocks of Power BI
- ✓ Fundamental Concepts of Power BI

Power BI Desktop

Learning Objectives

In this module you will be introduced to Power BI Desktop, software installation and overview of software

Topics Covered:

- ✓ Software Installation Procedures and Guidelines
- ✓ Overview of Software
- ✓ Power BI Sign UP

- ✓ Introduction to Tools and Terminologies
- ✓ Refreshing Power BI Service Data
- ✓ What is a Dashboard?
- ✓ What is a Report?
- ✓ How to share Dashboard and Report?

Data Source Connection and Data Transforming

Learning Objectives

In this module you will be introduced to how to connect to a data source, perform cleaning and transformation of data

Topics Covered:

- ✓ Connecting to a Data Source
- ✓ Upload a local CSV File
- ✓ Connect to Excel Data
- ✓ What is a Query Editor?
- ✓ Import Data vs Direct Query
- ✓ Data Cleaning and Data Transformation
- ✓ Merging and Appending

Hands-on

Datasets will be provided for practicing how to load and perform cleaning and data transformation.

Modelling using Power BI

Learning Objectives

In this module you will be introduced to data modelling, how to manage relationships, creating calculated columns, measures and quick measures

Topics Covered:

- ✓ Modelling of Data
- ✓ Manage Data Relationship
- ✓ Cardinality, Cross Filtering
- ✓ Default Summarization and Sort By
- ✓ Creating Calculated Columns
- ✓ Creating Quick Measures and Measures

Hands-on

In this you will be given hands-on to work on datasets, create calculated columns, manage relationships etc.,

Learning Objectives

In this module you will be introduced to DAX

Topics Covered:

- ✓ What is DAX?
- ✓ Why is DAX Important?
- ✓ DAX Syntax

- ✓ Data Types in DAX
- ✓ Functions in DAX
- ✓ Measures in DAX
- ✓ Operators in DAX
- ✓ Tables and Filtering in DAX
- ✓ Queries in DAX
- ✓ Parameter Naming in DAX

Hands-on

In this module hands-on will be given on various DAX functions

Modelling using Power BI

Learning Objectives

In this module you will be introduced to what is data visualization, importance of data visualization, creating charts using Power BI

Topics Covered:

- ✓ Creating Visualization
- ✓ Colour Formatting
- ✓ Sort Order
- ✓ Scatter plot and Bubble Charts
- ✓ Tooltips
- ✓ Slicers

- ✓ Cross Filtering, Highlighting
- ✓ Report Level Filters
- ✓ Drill Down and Drill Up
- ✓ Hierarchies
- ✓ Conditional Formatting, Tables, and Matrices
- ✓ KPI's
- ✓ Cards and Gauges
- ✓ Map Visualizations
- ✓ Custom Visuals
- ✓ Grouping and Binning
- ✓ Selection Pane, Bookmarks & Buttons
- ✓ Z-Order

Hands-on

In this module hands-on will be given on visualization, colour formatting, slicers, filters, grouping, selection pane etc.,

Reports and Dashboards

Learning Objectives

In this module you will be introduced to modify reports and dashboards, ask questions to Power BI Q&A

Topics Covered:

- ✓ Modify and Print a Report

- ✓ **Rename and delete report pages**
- ✓ **Add a filter to a page or report**
- ✓ **Set visualization interactions**
- ✓ **Print a report page**
- ✓ **Send a report to PowerPoint**
- ✓ **Create a Dashboard**
- ✓ **Create and manage dashboards**
- ✓ **Pin a report tile to a dashboard**
- ✓ **Pin a live report page to a dashboard**
- ✓ **Pin a tile from another dashboard**
- ✓ **Pin an Excel element to a dashboard**
- ✓ **Manage pinned elements in Excel**
- ✓ **Add a tile to a dashboard**
- ✓ **Build a dashboard with Quick Insights**
- ✓ **Set a Featured (default) dashboard**
- ✓ **Ask Questions about Your Data**
- ✓ **Ask a question with Power BI Q&A**
- ✓ **Tweak your dataset for Q&A**
- ✓ **Enable Cortana for Power BI**



Hands-on

In this module hands-on will be given on how to add filters to reports, set interactions, create dashboards etc.,

Publishing and Sharing Reports

Learning Objectives

In this module you will be introduced to how to publish and share your dashboards and reports, saving options.

Topics Covered:

- ✓ Introduction to Sharing Options
- ✓ How to Publish Report From Power BI Desktop?
- ✓ How to Publish Report to Web?
- ✓ How to Share Dashboards Using Power BI Service?
- ✓ What is Content Pack?
- ✓ How to Save as PDF?
- ✓ What is Row Level Security?
- ✓ How to Export Data from Visualization?

Hands-on

In this module hands-on will be given on how to share the dashboard with Power BI service, how to save as PDF and exporting data from the visualization.

POWER BI TRAINING

- ✓ **Module 1: Tableau Course Material (Duration – 5 Hours)**
 - Start Page
 - Show Me
 - Connecting to Excel Files
 - Connecting to Text Files
 - Connect to Microsoft SQL Server
 - Connecting to Microsoft Analysis Services
 - Creating and Removing Hierarchies
 - Bins
 - Joining Tables
 - Data Blending
- ✓ **Module 2: Learn Tableau Basic Reports (Duration – 5 Hours)**
 - Parameters
 - Grouping Example 1
 - Grouping Example 2
 - Edit Groups
 - Set
 - Combined Sets
 - Creating a First Report
 - Data Labels
 - Create Folders
 - Sorting Data



- Add Totals, Sub Totals, and Grand Totals to Report

✓ Module 3: Learn Tableau Charts (Duration – 4 Hours)

- Area Chart
- Bar Chart
- Box Plot
- Bubble Chart
- Bump Chart
- Bullet Graph
- Circle Views
- Dual Combination Chart
- Dual Lines Chart
- Funnel Chart
- Traditional Funnel Charts
- Gantt Chart
- Grouped Bar or Side by Side Bars Chart
- Heatmap
- Highlight Table
- Histogram
- Cumulative Histogram
- Line Chart
- Lollipop Chart
- Pareto Chart
- Pie Chart
- Scatter Plot



- Stacked Bar Chart
- Text Label
- TreeMap
- Word Cloud
- Waterfall Chart

✓ **Module 4: Learn Tableau Advanced Reports (Duration – 6 Hours)**

- Dual Axis Reports
- Blended Axis
- Individual Axis
- Add Reference Lines
- Reference Bands
- Reference Distributions
- Basic Maps
- Symbol Map
- Use Google Maps
- Mapbox Maps as a Background Map
- WMS Server Map as a Background Map

✓ **Module 5: Learn Tableau Calculations & Filters (Duration – 6 Hours)**

- Calculated Fields
- Basic Approach to Calculate Rank
- Advanced Approach to Calculate Ra
- Calculating Running Total

- Quick Filters
- Filters on Dimensions
- Conditional Filters
- Top and Bottom Filters
- Filters on Measures
- Context Filters
- Slicing Filters
- Data Source Filters
- Extract Filters
- Filters Introduction

✓ **Module 6: Learn Tableau Dashboards (Duration – 4 Hours)**

- Create a Dashboard
- Format Dashboard Layout
- Create a Device Preview of a Dashboard
- Create Filters on Dashboard
- Dashboard Objects
- Create a Story

✓ **Module 7: Server (Duration – 5 Hours)**

- Tableau online.
- Overview of Tableau Server.
- Publishing Tableau objects and scheduling/subscription.

DATA SCIENCE

(Four Projects with Resume Build Class)

Project Session Summary:

- 1 Credit Card Fraud Detection using Machine Learning in Python
- 2 Face Mask Detection using Convolutional Neural Network (CNN)
- 3 Fake News Detection using NLP Techniques
- 4 Emoji Predictions using RNN
- 5 Resume Modification & LinkedIn and Naukri Optimization

Credit Card Fraud Detection using Machine Learning in Python

Topics Summaries (Based on Project)

- ✓ Data Preprocessing: Handling missing values, scaling, and encoding categorical features
- ✓ Feature Engineering: Extracting key transaction patterns and reducing dimensions with PCA
- ✓ Anomaly Detection: Using Isolation Forests or Autoencoders for fraud identification.

- ✓ **Model Training:** Training supervised models like Random Forest or XGBoost on imbalanced data.
- ✓ **Evaluation Metrics:** Using Precision, Recall, F1-Score, and ROC-AUC for performance analysis.
- ✓ **Visualization:** Analyzing transaction patterns and fraud trends with Matplotlib/Seaborn.

Project Explanation

- ✓ **Project Overview:** Develop a machine learning model to detect fraudulent transactions in credit card data.
- ✓ **Dataset:** Utilize publicly available datasets such as the Kaggle Credit Card Fraud Detection dataset
- ✓ **Approach:** Preprocess the data, handle class imbalance (fraudulent transactions are rare), and train a classification model using techniques like Random Forest, XGBoost, and Isolation Forests.
- ✓ **Deployment:** Discuss how to deploy the model using Flask for real-time transaction monitoring.

Face Mask Detection using Convolutional Neural Network – CNN

Topics Summaries (Based on Project)

- ✓ **Data Preprocessing:** Resize, normalize, and augment images for training.
- ✓ **CNN Architecture:** Build or use pre-trained models like MobileNet or ResNet.

- ✓ **Training:** Optimize with Adam/Scheduler and use EarlyStopping for efficiency.
- ✓ **Evaluation:** Measure accuracy, precision, recall, F1-score, and ROC-AUC.
- ✓ **Real-Time Detection:** Use OpenCV with face detection for live mask detection.
- ✓ **Deployment:** Serve the model via Flask, FastAPI, or TensorFlow.js for browsers.
- ✓ **Edge Optimization:** Compress models using TensorFlow Lite or ONNX for devices.

Project Explanation

- ✓ **Project Overview:** Develop a deep learning model to detect face masks using CNN.
- ✓ **Dataset:** Use publicly available datasets such as the "Face Mask Detection" dataset.
- ✓ **Approach:** Preprocess the image data (resize, normalize), implement a CNN model with layers like convolution, pooling, and dense layers, and use pre-trained models like MobileNet for transfer learning
- ✓ **Deployment:** Discuss how to deploy the trained model using Flask for real-time detection, and optimize the model for edge devices using TensorFlow Lite.

Fake News Detection using NLP Techniques

Topics Summaries (Based on Project)

- ✓ **Data Preprocessing:** Clean text by removing stopwords, punctuation, and special characters.
- ✓ **Feature Extraction:** Use TF-IDF, Bag-of-Words, or word embeddings (Word2Vec, GloVe).
- ✓ **Text Representation:** Apply transformers like BERT for contextual embeddings.
- ✓ **Model Training:** Train classifiers like Logistic Regression, SVM, or LSTM models.
- ✓ **Evaluation:** Use accuracy, precision, recall, F1-score, and confusion matrices
- ✓ **Deployment:** Deploy using Flask, FastAPI, or Streamlit for real-time predictions.
- ✓ **Explainability:** Utilize SHAP or LIME for interpreting model predictions.

Project Explanation

- ✓ **Project Overview:** Build a machine learning model to classify fake news articles
- ✓ **Dataset:** Use datasets like the "Fake News Detection" dataset from Kaggle.
- ✓ **Approach:** Clean the textual data, extract features using methods like TF-IDF or word embeddings, and use machine learning algorithms such as Logistic Regression, SVM, or deep learning models like LSTM.
- ✓ **Deployment:** Deploy the model using Flask for real-time predictions, and integrate tools like SHAP or LIME to explain the model's decisions.

Emoji Predictions using RNN)

Topics Summaries (Based on Project)

- ✓ **Data Preprocessing:** Tokenize and clean text data for sequence modeling
- ✓ **Embedding Layer:** Convert words into dense vectors using Word2Vec or GloVe.
- ✓ **RNN Architecture:** Build models using LSTM or GRU layers for sequential data.
- ✓ **Training:** Optimize using Adam and handle overfitting with dropout layers.
- ✓ **Evaluation:** Measure accuracy and perplexity for sequence prediction
- ✓ **Real-Time Prediction:** Deploy with Flask or TensorFlow Lite for emoji suggestions
- ✓ **Attention Mechanism:** Integrate attention layers to improve contextual understanding.

Project Explanation

- ✓ **Project Overview:** Create a model that predicts the appropriate emoji for a given text input using RNN.
- ✓ **Dataset:** Use datasets like the "Emoji Prediction" dataset or create a custom dataset by pairing texts with corresponding emojis
- ✓ **Approach:** Tokenize the text data and use an embedding layer (Word2Vec or GloVe) followed by an LSTM or GRU model to predict the emoji.

- ✓ **Real-Time Prediction:** Discuss how to deploy the model using Flask or TensorFlow Lite for realtime emoji prediction.

Resume Modification

- ✓ **For Freshers:**
 - Highlight key skills (Machine Learning, NLP, Deep Learning).
 - Include project experience with bullet points explaining responsibilities and achievements.
- ✓ **For Non-IT to Java Developer Transition:**
 - Focus on transferable skills (problem-solving, logic, teamwork).
 - Highlight certifications and Data Science projects completed during training
- ✓ **For Experienced Data Scientists:**
 - Emphasize achievements like application performance improvements, cloud migrations, and team leadership.
- ✓ **LinkedIn:**
 - Write a compelling summary and add skills like Python programming, data manipulation with NumPy and Pandas, machine learning (supervised/unsupervised), deep learning (CNNs, RNNs), NLP, data visualization, model deployment, APIs, cloud platforms, big data technologies.

- Showcase projects and certifications. o Engage by sharing articles and commenting on relevant topics.

✓ Naukri:

- Complete your profile with detailed work experience and skills.
- Use keywords such as Python programming, data manipulation with NumPy and Pandas, machine learning (supervised/unsupervised), deep learning (CNNs, RNNs), NLP, data visualization, model deployment, APIs, cloud platforms, big data technologies
- Frequently update the profile for better visibility.



Advanced Certification In Generative AI & Machine Learning



Master Generative AI and Redefine Your
Career Path

ADVANCED MACHINE LEARNING & GENERATIVE AI WITH TRANSFORMERS, LLMS, AND PROMPT ENGINEERING

Foundation in NLP & Deep Learning

- ✓ Word Embeddings: Word2Vec, GloVe, FastText
- ✓ Sequence Models: RNNs, GRUs, LSTMs
- ✓ Attention Mechanisms & Encoder-Decoder Frameworks
- ✓ Preprocessing pipelines and tokenization techniques

Hands-on Labs:

- ✓ Train LSTM for text generation
- ✓ Word2Vec visualization on custom corpus

Transformers from Scratch

- ✓ Self-attention, scaled dot-product attention
- ✓ Multi-head attention, positional encoding
- ✓ Encoder, decoder stack
- ✓ Residual connections and layer normalization

Hands-on Labs:

- ✓ Implement Transformer block in PyTorch
- ✓ Visualize attention maps on a sentence

Hugging Face Transformers

- ✓ Architecture and tokenizer usage
- ✓ Model loading, inference, and fine-tuning
- ✓ BERT, RoBERTa, T5, DistilBERT, GPT-2
- ✓ Dataset loading and batching with Datasets

Hands-on Labs:

- ✓ Text classification with BERT
- ✓ Summarization with T5
- ✓ Q&A using DistilBERT

Large Language Models (LLMs) In-Depth

- ✓ Architecture of GPT-3, GPT-4, LLaMA, PaLM
- ✓ Training data pipelines and tokenizers
- ✓ Zero-shot, few-shot, and in-context learning
- ✓ Model scaling laws and performance metrics

Hands-on Labs:

- ✓ GPT-3.5 API integration
- ✓ Few-shot prompting with LLaMA (via Hugging Face)
- ✓ Generate domain-specific text (e.g., finance, education)

Text Generation & Decoding Strategies

- ✓ Greedy, Beam Search, Top-k, Nucleus Sampling
- ✓ Controlled text generation
- ✓ Token probabilities and temperature tuning
- ✓ Limitations (hallucinations, verbosity)

Hands-on Labs:

- ✓ Implement decoding strategies manually
- ✓ Compare output quality with different settings

Prompt Engineering & Reasoning Frameworks

- ✓ Prompt design patterns: QA, summarization, classification
- ✓ Instruction tuning, system prompts
- ✓ Chain-of-thought prompting, ReAct prompting
- ✓ Prompt injection vulnerabilities and mitigation

Hands-on Labs:

- ✓ Build a prompt library
- ✓ CoT for logic/mathematical reasoning
- ✓ Prompt attack simulations and defenses

Fine-Tuning & Parameter Efficient Training

- ✓ Traditional fine-tuning vs. Adapter/LoRA
- ✓ QLoRA for memory-efficient tuning
- ✓ Tokenization, special tokens, padding

- ✓ Evaluation and loss visualization

Hands-on Labs:

- ✓ Fine-tune BERT on sentiment analysis
- ✓ QLoRA fine-tuning on Alpaca or LLaMA
- ✓ Visualize embedding shifts pre/post fine-tuning

Retrieval-Augmented Generation (RAG)

- ✓ Concept of RAG: retrieval + generation
- ✓ Document chunking, vector embedding
- ✓ Similarity search with FAISS/Chroma
- ✓ Indexing PDFs, websites, and text files

Hands-on Labs:

- ✓ Create a chatbot with LangChain + OpenAI + FAISS
- ✓ QLoRA fine-tuning on Alpaca or LLaMA
- ✓ Visualize embedding shifts pre/post fine-tuning

Building GenAI-Powered Applications

- ✓ Use LangChain agents and tools
- ✓ Create dynamic workflows with chains
- ✓ Streamlit or Gradio for LLM UI
- ✓ Using LLMs with voice, images, or tabular data

Hands-on Labs:

- ✓ Resume analysis bot
- ✓ Virtual mentor for Python programming
- ✓ Legal contract summarizer

Ethics, Bias, and Safe Deployment

- ✓ Understanding AI alignment and hallucinations
- ✓ Bias in training data and fairness metrics
- ✓ Moderation, red-teaming, adversarial inputs
- ✓ Transparency and auditability of outputs

Hands-on Labs:

- ✓ Identify bias in LLM-generated outputs
- ✓ Apply moderation filters using OpenAI tools
- ✓ Red-team testing with adversarial prompts

Capstone Project & Presentation

- ✓ Choose a domain: Healthcare, Education, Finance, HR, Law, etc
- ✓ Define a real-world GenAI use-case
- ✓ Build full-stack solution (backend + LLM + UI)
- ✓ Present to panel with demo and documentation